The compact F_4 sigma-MGF on V_{26}

Proof and exact low-degree matrix verification

Computational proof companion

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Abstract

For the trace-zero Albert module we prove the compact Molien–Weyl expression, construct the 26-dimensional torus matrices, and exactly verify $\mathcal{S}_{F_4,26} = 1 + h_2 + h_3 + O_{\geq 4}$. The one-copy Hilbert series is $((1-t^2)(1-t^3))^{-1}$.

1 Formula and proof

The Cauchy identity followed by Haar projection gives

$$\boxed{\mathcal{S}_{K,V} = \int_K \prod_a \det(I - t_a \rho(g))^{-1} dg = \sum_\lambda \dim(\mathbb{S}_\lambda V)^K s_\lambda(\mathbf{t}).}$$

Equivalently, direct determinant expansion proves $[m_\tau]\mathcal{S}_{K,V} = \dim(\bigotimes_i \operatorname{Sym}^{\tau_i} V)^K$. Weyl integration turns this into

$$\frac{1}{|W|} \operatorname{CT}_z \left[\prod_{\alpha>0} (1 - z^\alpha)(1 - z^{-\alpha}) \prod_{a,\mu} (1 - t_a z^\mu)^{-m(\mu)} \right].$$

For V_{26} the weights are the 24 short roots and zero with multiplicity two; $|W(F_4)| = 1152$. Therefore

$$\boxed{\mathcal{S}_{F_4,26} = \frac{1}{1152} \operatorname{CT}_z \left[D_{F_4}(z) \prod_a (1 - t_a)^{-2} \prod_{\beta \in \Phi_{\text{short}}} (1 - t_a z^\beta)^{-1} \right].}$$

2 Invariant tensors

The trace-zero Albert algebra carries the restricted trace form and the restricted determinant. These are respectively a unique symmetric quadratic and a unique symmetric cubic invariant. No other Schur type occurs below degree four, so

$$\boxed{\mathcal{S}_{F_4,26} = 1 + h_2 + h_3 + O_{\geq 4}.}$$

The standard invariant theory of the trace-zero Albert algebra gives $\mathbb{C}[V_{26}]^{F_4} = \mathbb{C}[q, c]$, with generators of degrees two and three. Therefore $\mathcal{S}(t, 0, \ldots) = ((1-t^2)(1-t^3))^{-1}$. This all-degree invariant-ring description is an external input; the notebook verifies only the displayed low-degree terms.

3 Verification method and evidence

Starting with the F_4 Cartan matrix, the notebook reflects a short simple root to obtain 24 short roots and appends two zero weights. It constructs

$$M(\theta) = \operatorname{diag}(e^{i\langle \mu, \theta \rangle})_\mu \in U(26)$$

and checks the determinant integrand against its eigenvalue product. Schur characters are computed independently by Jacobi–Trudi. Their Haar integrals are exact: for a character f invariant under W,

$$\frac{1}{|W|}\,\mathrm{CT}(D_{F_4}f) = \mathrm{CT}\left(f\prod_{\alpha>0}(1-z^{-\alpha})\right),$$

and the Weyl-denominator identity supplies every coefficient on the right. The direct monomial-basis check returns

$$([m_{(2)}],[m_{(1,1)}],[m_{(3)}],[m_{(2,1)}],[m_{(1,1,1)}]) = (1,1,1,1,1).$$

λ	$\varnothing$	(1)	(2)	$(1,1)$	(3)	$(2,1)$	$(1,1,1)$
proposed	1	0	1	0	1	0	0
exact Weyl check	1	0	1	0	1	0	0

The generated root count is 48, the matrix dimension is 26, the unitarity residual is 2.8×10^{-16}, and the two integrand evaluations agree to 1.3×10^{-14}. Every degree-at-most-three coefficient passes.

Reproducibility. Run `f4_verification.py` with marimo.

External invariant-theory source

See T. A. Springer and F. D. Veldkamp, *Octonions, Jordan Algebras and Exceptional Groups*, Springer Monographs in Mathematics (2000), for the Albert-algebra realization and its norm invariants.